

```
scala> import org.apache.spark.sql.SparkSession
import org.apache.spark.sql.functions._

val spark = SparkSession.builder
  .appName("TextCleaningStream")
  .master("local[*]")
  .getOrCreate()

import spark.implicits._
warning: 1 deprecation (since 2.13.3); for details, enable `:setting -deprecation` or `:replay -deprecation`
26/05/02 13:44:58 WARN SparkSession: Using an existing Spark session; only runtime SQL configurations will take effect.
import org.apache.spark.sql.SparkSession
import org.apache.spark.sql.functions._
val spark: org.apache.spark.sql.SparkSession = org.apache.spark.sql.classic.SparkSession@6f6c01d9
import spark.implicits._

scala> val lines = spark.readStream
  .format("socket")
  .option("host", "localhost")
  .option("port", 9999)
  .load()
26/05/02 13:45:10 WARN TextSocketSourceProvider: The socket source should not be used for production applications! It does not support recovery.
val lines: org.apache.spark.sql.DataFrame = [value: string]

scala> val words = lines.as[String]
val words: org.apache.spark.sql.Dataset[String] = [value: string]
```

```
scala> val stopWords = Seq("is", "the", "a", "an", "and", "to")

val cleaned = words.map { line =>
  val lower = line.toLowerCase
  val noSpace = lower.trim.replaceAll("\\s+", " ")
  val noPunct = noSpace.replaceAll("[^a-zA-Z ]", "")

  val filtered = noPunct.split(" ")
    .filter(word => word.nonEmpty && !stopWords.contains(word))

  val lemmatized = filtered.map {
    case w if w.endsWith("ing") => w.dropRight(3)
    case w if w.endsWith("ed") => w.dropRight(2)
    case w => w
  }

  lemmatized.mkString(" ")
}

val stopWords: Seq[String] = List(is, the, a, an, and, to)
val cleaned: org.apache.spark.sql.Dataset[String] = [value: string]
```

```

nidhi@NidhiDB:~$ sudo apt install netcat-openbsd -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
netcat-openbsd is already the newest version (1.226-1ubuntu2).
netcat-openbsd set to manually installed.
The following packages were automatically installed and are no longer required:
  libllvm19 libwayland-server0
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
nidhi@NidhiDB:~$ nc -lk 9999
hello spark streaming test
spark is working now

```

```

scala> val query = cleaned.writeStream
      |   .outputMode("append")
      |   .format("console")
      |   .start()
26/05/02 13:50:52 WARN ResolveWriteToStream: Temporary checkpoint location created which is deleted normally when the query didn't fail: /tmp/temporary-d2177d16-8363-420f-bf8e-6ada3574b0a0. If it's required to delete it under any circumstances, please set spark.sql.streaming.forceDeleteTempCheckpointLocation to true. Important to know deleting temp checkpoint folder is best effort.
26/05/02 13:50:52 WARN ResolveWriteToStream: spark.sql.adaptive.enabled is not supported in streaming DataFrames/Datasets and will be disabled.
val query: org.apache.spark.sql.streaming.StreamingQuery = org.apache.spark.sql.execution.streaming.runtime.StreamingQueryWrapper@413eb0a6

[Stage 8:>                                                                    (0 + 12) /

-----
Batch: 0
-----
+-----+
|value|
+-----+
+-----+

-----
Batch: 1
-----
+-----+
|           value|
+-----+
|hello spark strea...|
+-----+

-----
Batch: 2
-----
+-----+
|           value|
+-----+

```